

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

Atomic Physics

Q1

A H-atom in ground state has time period $T = 1.6 \times 10^{-16}$ sec. find the frequency of electron in first excited state

एक H-परमाणु के इलेक्ट्रॉन का निम्नतम अवस्था में आवर्तकाल $T = 1.6 \times 10^{-16}$ sec. है। तब इलेक्ट्रॉन की प्रथम उत्तेजित अवस्था में आवर्ती होगी –

- (1) 7.8×10^{14} (2) 7.8×10^{16} (3) 3.7×10^{14} (4) 3.7×10^{16}

7th Jan Morning

Sol

(1)

$$T \propto \frac{r}{v} \propto \frac{n^2}{Z} \times \frac{n}{Z} \propto \frac{n^3}{Z^2}$$

$$\frac{T_1}{T_2} = \frac{n_1^3}{n_2^3} = \frac{1}{8}$$

$$T_2 = 8T_1 = 8 \times 1.6 \times 10^{-16} = 12.8 \times 10^{-16}$$

$$f_2 = \frac{1}{12.8 \times 10^{-16}} \approx 7.8 \times 10^{14}$$

Q2

An electron & a photon have same energy E. Find the ratio of de Broglie wavelength of electron to wavelength of photon. Given mass of electron is m & speed of light is C.

एक इलेक्ट्रॉन तथा एक फोटोन की ऊर्जा एकसमान है। इलेक्ट्रॉन तथा फोटोन की डी-ब्रोग्ली तरंगदैर्घ्य का अनुपात ज्ञात कीजिए। दिया गया है, इलेक्ट्रॉन का द्रव्यमान = m प्रकाश की चाल = C

- (1) $\frac{1}{C} \left(\frac{E}{2m} \right)^{1/2}$ (2) $\left(\frac{E}{m} \right)^{1/2} C$ (3) $\frac{\sqrt{2mE}}{C}$ (4) $\left(\frac{E}{2m} \right)^{1/2}$

7th Jan Evening

Sol

(1)

$$\lambda_d \text{ for electron} = \frac{h}{\sqrt{2mE}}$$

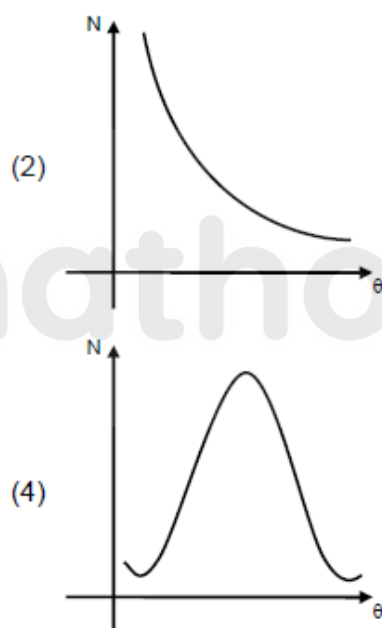
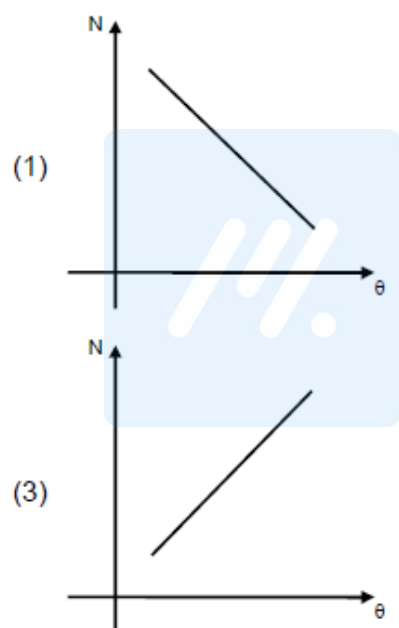
$$\lambda \text{ for photon} = \frac{hc}{E}$$

$$\text{Ratio} = \frac{h}{\sqrt{2mE}} \cdot \frac{E}{hc} = \frac{1}{c} \sqrt{\frac{E}{2m}}$$

Q3

Number N of the α -particles deflected in Rutherford's α -scattering experiment varies with the angle of deflection θ . Then the graph between the two is best represented by.

रदरफोर्ड के α -प्रकीर्णन प्रयोग में विचलित होने वाले α -कणों की संख्या N विचलन कोण θ के साथ बदलती है तो इन दोनों के बीच सर्वोत्तम आरेख होगा।

8th Jan Morning

Sol

(2)

$$N \propto \frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$$

Q4

In H-spectrum wavelength of 1st line of Balmer series is $\lambda = 6561\text{\AA}$. Find out wavelength of 2nd line of same series in nm.

8th Jan Evening

Sol

$$\begin{aligned}\frac{1}{\lambda} &= RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \\ \frac{1}{\lambda_1} &= R(1)^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5R}{36} \\ \frac{1}{\lambda_2} &= R(1)^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{3R}{16} \\ \frac{\lambda_2}{\lambda_1} &= \frac{20}{27} \\ \lambda_2 &= \frac{20}{27} \times 6561\text{\AA} = 4860\text{\AA} \\ &= 486\text{ nm}\end{aligned}$$

Q5

Kinetic energy of the particle is E and it's De-Broglie wavelength is λ . On increasing it's KE by ΔE , it's new De-Broglie wavelength becomes $\frac{\lambda}{2}$. Then ΔE is

एक कण जिसकी गतिज ऊर्जा E तथा डी-ब्रोग्ली तरंगदैर्घ्य λ है। इसकी गतिज ऊर्जा में ΔE की वृद्धि करने पर उसकी नयी डी-ब्रोग्ली तरंगदैर्घ्य $\frac{\lambda}{2}$ हो जाती है। तब ΔE होगा—

- (1) 3E (2) E (3) 2E (4) 4E

9th Jan Morning

Sol

(1)

$$\lambda = \frac{h}{\sqrt{2(KE)m}} \Rightarrow \lambda \propto \frac{1}{\sqrt{KE}}$$

$$\frac{\lambda}{\lambda/2} = \sqrt{\frac{KE_f}{KE_i}}$$

$$4KE_i = KE_f$$

$$\Rightarrow \Delta E = 4KE_i - KE_i = 3KE = 3E$$

Q6

Consider an infinitely long current carrying cylindrical straight wire having radius 'a'. Then the ratio of magnetic field at distance $\frac{a}{3}$ and $2a$ from axis of wire is.

एक अनन्त लम्बे धारावाही बेलनाकार सीधे तार कि त्रिज्या 'a' है। तब अक्ष से $\frac{a}{3}$ तथा $2a$ दूरी पर चुम्बकीय क्षेत्र का अनुपात होगा—

(1) $\frac{3}{5}$

(2) $\frac{2}{3}$

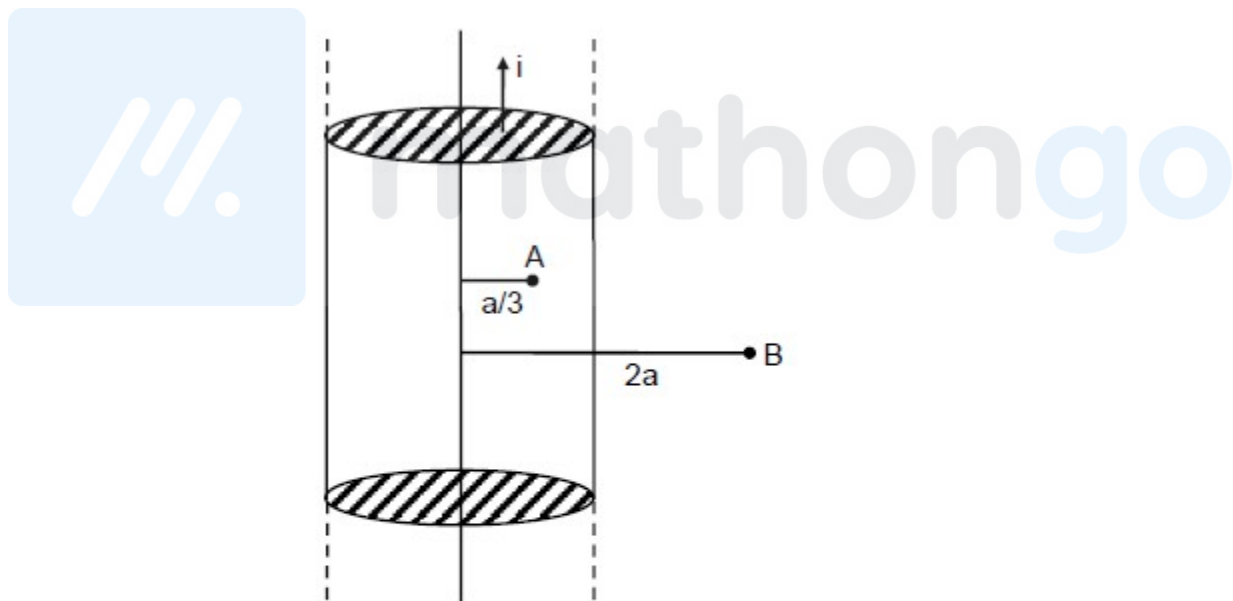
(3) $\frac{1}{2}$

(4) $\frac{4}{3}$

9th Jan Evening

Sol

(2)



$$B_A = \frac{\mu_0 i r}{2\pi a^2} = \frac{\mu_0 i \frac{a}{3}}{2\pi a^2} = \frac{\mu_0 i}{\pi a^2} \frac{a}{6} = \frac{\mu_0 i}{6\pi a}$$

$$B_B = \frac{\mu_0 i}{2\pi(2a)}$$

$$\frac{B_A}{B_B} = \frac{4}{6} = \frac{2}{3}$$

Q7

Atomic Physics**JEE Mains 2020 Jan Chapter wise Question Bank**

An electron $(-|e|, m)$ is released in Electric field E from rest. Rate of change of de-Broglie wavelength with time will be.

एक इलेक्ट्रॉन $(-|e|, m)$ विद्युत क्षेत्र E में स्थिर अवस्था से छोड़ा जाता है, तो डी-ब्रोग्ली तरंगदैर्घ्य के समय के साथ परिवर्तन की दर होगी—

(1) $-\frac{h}{2|e|}$

(2) $-\frac{h}{2|e|t}$

(3) $-\frac{h}{|e|Et^2}$

(4) $-\frac{2ht^2}{|e|E}$

9th Jan Evening

Sol

(3)

$$\lambda_D = \frac{h}{mv}$$

$$\therefore v = at$$

$$v = \frac{eE}{m}t \quad (a = \frac{eE}{m})$$

$$\lambda_D = \frac{h}{m\left(\frac{eE}{m}\right)t}$$

$$\lambda_D = \frac{h}{eEt}$$

$$\frac{d\lambda_d}{dt} = -\frac{h}{|e|Et^2}$$

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